

WASP Software Engineering Course Assignment 2026

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Build Instructions

To build the PDF, run

```
pandoc assignment.md \  
  --pdf-engine typst \  
  -M author="Ziyong Cheong, Chalmers" \  
  -M title="WASP Software Engineering Course Assignment 2026" \  
  -C --bibliography=assignment.bib \  
  -V mainfont="Times New Roman" \  
  -V fontsize=11pt \  
  -V linkcolor=337ab7 \  
  -o assignment.pdf
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This requires [Pandoc](#), [Typst](#), and the Times New Roman font to be installed.

Introduction

I work in the field of theoretical machine learning, studying the generalisation of large language models (LLMs) using tools from information theory. The key intuition here is that prediction and compression represent two sides of the same coin: To optimally predict, e.g., the next token in a sequence, one has to know the true joint distribution of all possible tokens (to compute the conditional probability). At the same time, knowing the true joint distribution is also what is necessary to compress a given sequence using, e.g., Huffman trees. There is also empirical evidence supporting this connection in the form of knowledge distillation, with distilled models such as TinyBERT retaining over 95% of the base model's performance at a fraction of the size (Jiao et al. 2020).

The approach that is taken in my research is inspired by the field of representation learning. We conjecture that LLMs learn a compressed representation of the input which is minimally sufficient, i.e., it retains the minimal amount of information from the input which suffices to predict the output. We also conjecture that this compression is not only information theoretic, but geometric in nature, in other words, the representations form clusters.

Lecture Principles

One of the two principles that I took away from the main lectures was to adopt an engineering mindset and prioritise system reliability and ease of scale, as well as thinking more long-term when writing model code to empirically validate theorems. Concretely for my research, this entails

- properly documenting the origins of particular functions in anticipation of future me reusing the code,
- writing modular code to facilitate tweaking the model, and
- using git extensively to preserve the history of changes.

The other concept was the difference between accidental and essential complexity. Accidental complexity is complexity that arises from the tooling used to solve a problem, and induces friction when working, whereas essential complexity is the inherent complexity of the task at hand. Reducing accidental complexity by learning or using better tools is thus only meaningful up to a point and has diminishing returns. For example, I should not spend too much time on setting up my LaTeX configuration and learning the inner details of LaTeX.

Guest-Lecture Principles

Two concepts I learnt from the guest lectures are *requirements engineering* and *cognitive intensity balance*. With regards to requirements engineering, it's good to occasionally take a step back and reevaluate what the exact requirements of my research are: What needs to be proven? Who would benefit from any new results? What is the main goal of my research? The pipeline presented in the guest lecture of considering

1. Who are the stakeholders?
2. What are their goals?
3. How can the stakeholders' goals be formulated as a general vision for the project?
4. How can this general vision be broken down into concrete requirements? seems like a useful and practical way to come up with these requirements.

The second concept was that cognitive load should be well balanced. Around 75% of all tasks should have low cognitive load, while 20% of tasks should have medium cognitive load, and 5% should have high cognitive load. This results in a good level of motivation. On the flipside, automating away most of the easy tasks leads to mental fatigue and stress.

Data Scientists, Software Engineers, and AI Engineers

The book (Kästner 2025) describes data scientists as being more model centric, meaning they tend to only focus on the machine learning (ML) model used, and ignore surrounding factors such as the latency and cost of inference. They are also described as taking a more scientific approach to work, and not enjoying the 'grunt' engineering work of tweaking infrastructure. The role of the data scientist in a company is thus to research and develop the core ML model. In contrast, software engineers, as the name suggests, take up the engineering role and handle the other components in the system. They are also described as having a limited interest in ML, leading to a naïve approach to ML modelling.

I find these descriptions to be, as self-admitted in the book, rather superficial. While there is certainly a split between research and development, I think this split is along task lines, and not within the team itself.

The book also advocates for a system-wide view, as opposed to the model-centric view data scientists allegedly have. This is difficult to apply to my research, seeing as I am not dealing with production ML. However, one possible application is to also look at the data when deriving generalisation bounds for ML models. This is also the view expressed in (Zhang et al. 2021) which empirically showed how a data-agnostic view of model generalisation is insufficient.

I think the (oversimplified) roles of data scientist and software engineer as described by Kästner (2025) will remain distinct in the future, but their demarcation line will blur. Since the only way in current society to capitalise on new models is to found a startup or join a big tech company, developers will still specialise as data scientists and software engineers, and it is the interaction between the two roles which will blur the line. AI engineering, defined to be software engineering for AI, will play a small role in this, as

- foundation models offload most of the research work and modularise the model,
- AI coding assistants expose software engineers to ML models and allow data scientists to write better code, and
- more AI companies emerge, bringing more data scientists and software engineers in contact with each other.

Paper Analysis

Paper 1

For the first paper, I looked at (Ferreira et al. 2026). The core idea of this paper was to optimise ML models for both energy efficiency as well as performance. Ferreira et al. (2026) introduced the energy consumption optimiser (ECOpt) which uses multi-objective Bayesian optimisation (MOBO) to find trade-offs between the two metrics. In particular, ECOpt

- constructs a Pareto frontier between energy efficiency and model performance for different hyperparameter configurations,
- using MOBO which automatically optimises multiple metrics simultaneously,
- considers both training and inference energy consumption, since these can differ drastically, and
- does not simply measure FLOPs and parameter count, since these can be uncorrelated to energy use, and are thus not necessarily good proxies.

These ideas are directly related to AI engineering. Reducing the energy required during inference can directly reduce operational costs, particularly when a model is deployed at scale. The paper therefore illustrates how model selection should consider system-level properties rather than only model performance.

Taking the transcription service example from (Kästner 2025), a smaller less energy-intensive model could be used, if its performance suffices. This would reduce energy costs of inference and allow the service to possibly run on cheaper hardware, reducing operational costs. In addition, since the model selection is automatically optimised by MOBO, this frees up the developers' time to work on other parts of the system, such as the payment service and website, or to fix bugs.

This is relevant to my own research since I frequently need to train multiple models to empirically validate theoretical claims. By choosing less energy-intensive (but still sufficiently performant) models, I would have an easier time performing large-scale empirical validation. The paper also lead to my self reflection about the purpose of the experiments; If the goal is to merely test the validity of theorems, it is unnecessary to use the largest, best performing model. Using the largest model for experiments might in fact be a hindrance to evaluations of future theorems.

Paper 2

The second paper (Ren et al. 2024) investigates the relationship between general model capabilities and safety. The main result is that general performance and safety are sometimes, but not always, (negatively) correlated. This suggests that simply increasing model capability does not necessarily lead to a safer model. More precisely, performance on many popular ML safety benchmarks is directly correlated to model scale. However, only some actual safety tasks improve with scale, e.g., adversarial robustness, machine ethics, truthfulness, while some (sycophancy, bias, discrimination) do not. The authors identify a so-called *capabilities component*, i.e., the eigenvector corresponding to the largest eigenvalue of the correlation matrix $C = A^\top A$ between safety tasks, where A_{ij} is the score of model i on benchmark j . It is this capabilities component which is correlated with model size.

The connection of these results to AI engineering is obvious: The model with the highest performance is not always the safest model. In fact, the most performant model might even be *less* safe than other models. An AI engineer thus has to not only consider which model to use, but also if the tests used to evaluate said model provide meaningful insight on the model's behaviour.

One possible application is the evaluation of a customer-service chatbot. Suppose a new model obtains a higher score on a safety benchmark. Before concluding that the new model is safer, the engineering team should determine whether the benchmark is actually sensitive to the particular safety properties that matter for the chatbot. They could supplement the benchmark with application-specific tests, for example testing whether the model reveals private information, follows inappropriate instructions, or produces harmful responses. This would make the evaluation more closely aligned with the actual requirements of the system.

The paper also has an interesting connection to my own research. I study generalisation of ML models, and the paper highlights the importance of being precise about what a benchmark or theoretical quantity actually measures. In fact, studying which exact metric appears in the generalisation bound reveals interesting qualities about the model being studied. For example, a generalisation bound with an information-theoretic metric would imply that the model is performing some form of compression.

Evidence, Originality, and GenAI-Use

Five sources were used for this report. This section discusses each source used, in order of their appearance in the [References](#). The first source is a conference paper from CAIN 2026 which, despite only being a B-tier conference, is double-blind peer reviewed, and thus, decently trustworthy. The second source is a conference paper from EMNLP 2020 which is a highly prestigious A* conference with double-blind peer review. The third source is the book by Kästner (2025) published by MIT press. The author is an associate professor at Carnegie Mellon, an established university. The fourth source is a conference paper from NeurIPS 2024, the most prestigious ML conference. The final source is a seminal paper in my field of research. No generative AI was used for the production of this report.

References

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